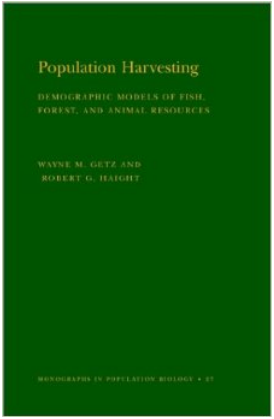
Harvest Models



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Learning objectives

Sustainable harvest and geometric growth

Sustainable harvest and logistic growth

Definition of maximum sustainable yield (MSY)

Limitations of MSY

Additive vs compensatory mortality

A sustainable (and large) harvest is a common objective in game management

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Sustainable harvest: A harvest that is balanced by population growth such that $N_{t+1} = N_t$

Harvest and geometric growth

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Example when $K = 1000$ and $r_{max} = 0.1$

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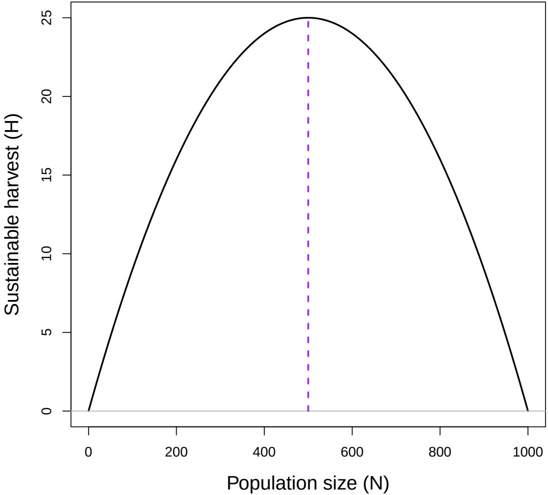

Example when $K = 1000$ and $r_{max} = 0.5$

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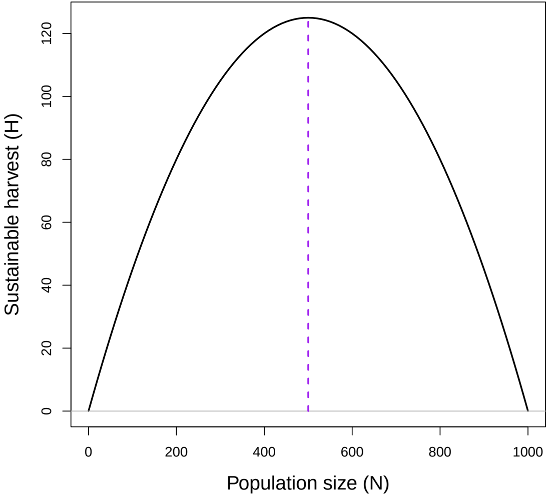

- MSY is found when $N = K/2$
- The actual maximum yield is $H = r_{max}K/4$
- The optimal harvest rate is $h = r_{max}/2$

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- Evolutionary consequences?

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- If harvest is compensated for by improved survival, harvest is a form of **compensatory mortality**
- However, if harvest is not compensated for by improved survival, harvest is a form of **additive mortality**

If harvest mortality is additive, extra caution is needed to ensure that harvest doesn't cause long-term population declines.

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Compensatory mortality example

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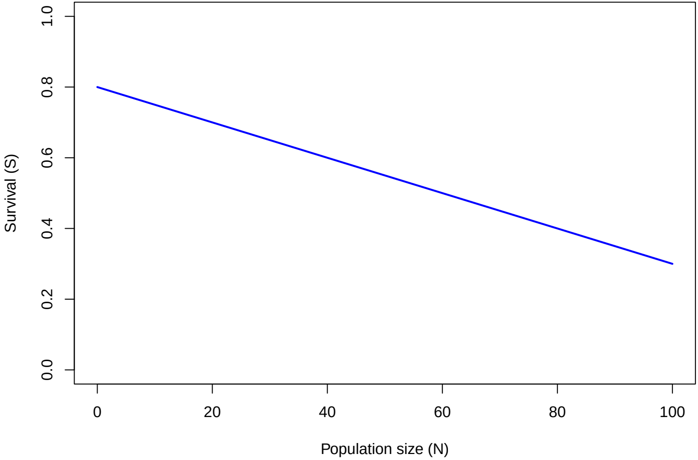
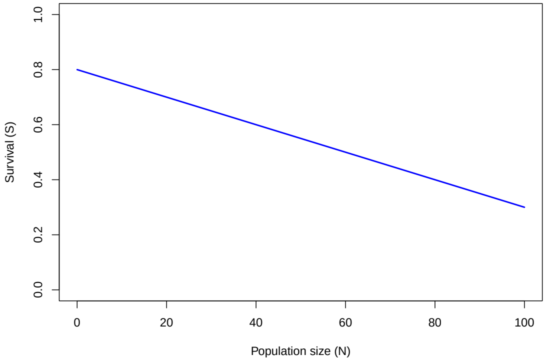


$S = 0.8 - 0.005 \times N$

- Suppose 20 individuals are harvested from an initial population of 100 individuals.



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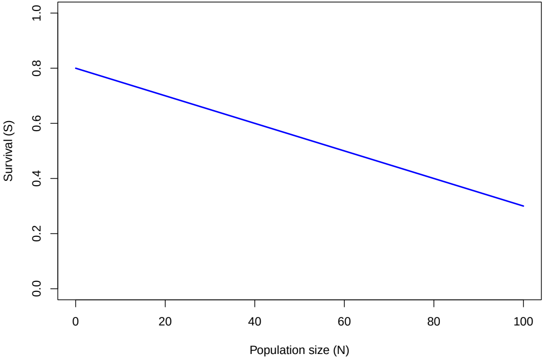

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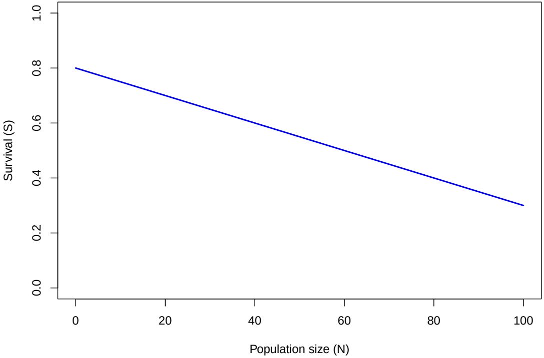
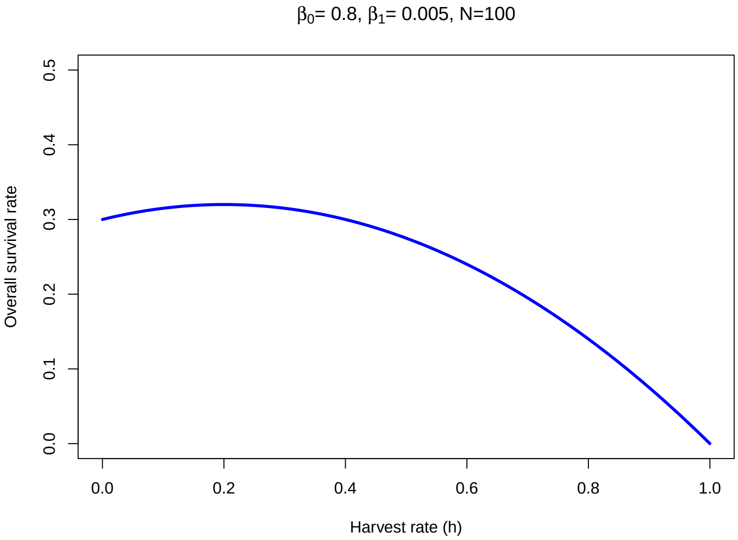
- Suppose 20 individuals are harvested from an initial population of 100 individuals.
- How many individuals will remain at the end of the year?
- How many would have remained at the end of the year if no hunting had occurred?



The overall survival rate ($\bar{S}$) is product of survival throughout the hunting season $(1-h)$ and survial after the hunting season

$$\bar{S} = (1-h)(\beta_0 - \beta_1(N - Nh))$$

Overall survival vs. harvest rate



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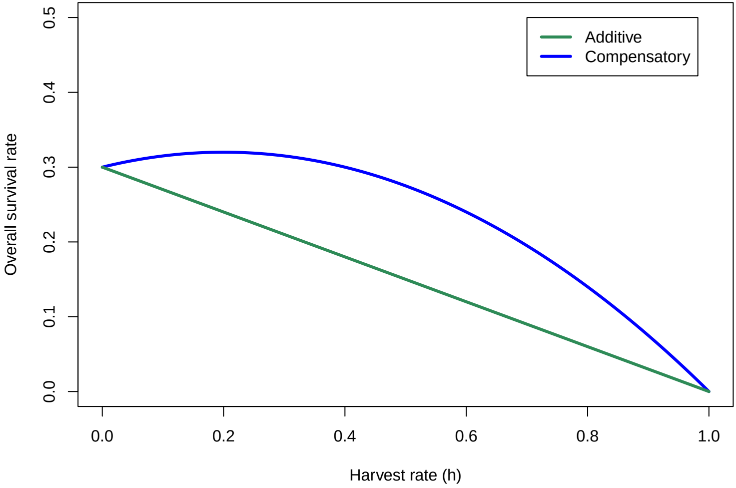
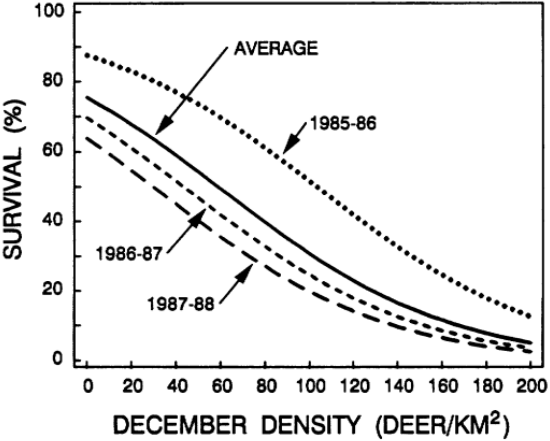


Mule deer fawn survival (From Bartman et al. 1992)



Key points

- If growth is geometric, sustainable harvest occurs when $h = r$.
- If growth is logisitic, maximum sustainable yield occurs at $N = K/2$.
- If survival is density-dependent, harvest mortality can be compensated for by increased survival of remaining individuals (up to a point).
- If mortality is additive, extra caution is needed because harvest is adding to natural mortality without any compensation.
- Managers need to know if harvest mortality is additive or compensatory when setting harvest regulations.

Read pages 22–25 in Conroy and Carroll